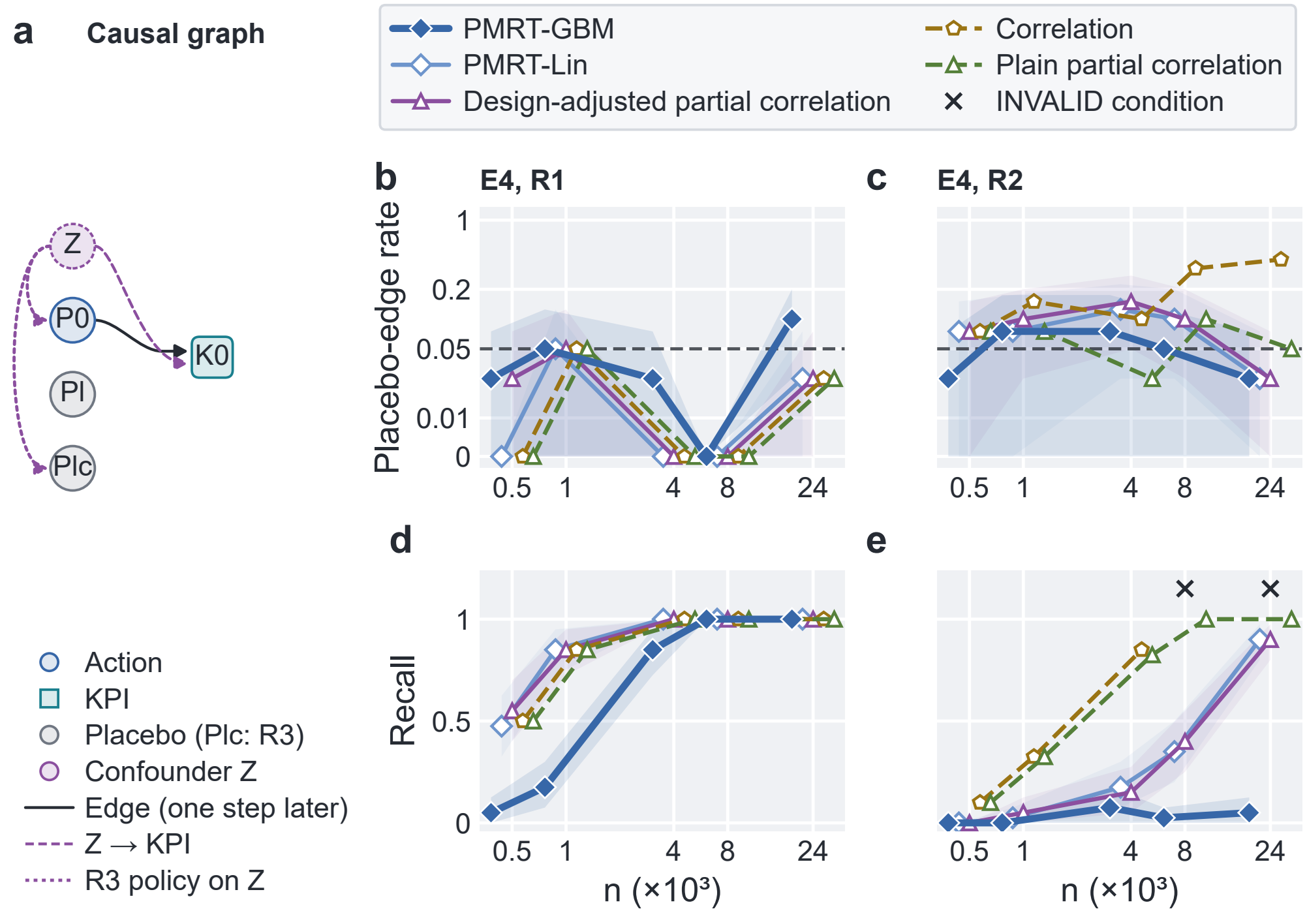
a
Causal graph
PMRT-GBM
PMRT-Lin
Design-adjusted partial correlation
Correlation
Plain partial correlation
INVALID condition
b
E4, R1
c
E4, R2
Placebo-edge rate
1
0.2
0.05
0.01
0
0.5
1
4
8
24
Z
P0
K0
PI
Plc
d
e
Recall
1
0.5
0
n (×10³)
Action
KPI
Placebo (Plc: R3)
Confounder Z
Edge (one step later)
Z → KPI
R3 policy on Z